

Caffeine Algorithm — Cognitive State Model for Realistic Agents

Julien Pierre Salomon

Status: Research / Formalisation

Claim: Insert a continuous-time Cognitive State Filter between intent and expression to capture bounded rationality and biologically grounded constraints.

1 The Caffeine Algorithm: A Unified Mathematical Model of Circadian-Modulated Cognitive State for Synthetic Agents

2 Epistemological Foundations of the Cognitive State Filter

The current paradigm of autonomous agent architecture is predominantly driven by direct intent-to-action mapping. In this “stateless” or “always-on” model, an agent’s output is a direct function of its prompt instructions and the immediate context window. While computationally efficient, this approach fails to capture the bounded rationality and biological constraints that characterize human cognition.

Human behaviour is not merely a sequence of logical deductions; it is the product of a dynamic interplay between internal homeostatic drives, circadian rhythms, and neurochemical modulations. The Caffeine Algorithm proposes a paradigm shift: the insertion of a continuous-time Cognitive State Filter between intent and expression.

This report formalises a mathematical model where an agent’s observable behaviour—its timing, tone, depth, and error rate—is modulated by a low-dimensional state vector evolving in time. This vector, $S(t)$, represents the “metabolic” and “emotional” substrate of the agent. By simulating the constraints of energy, patience, and arousal, we move from agents that simply act to agents that behave with ecological validity.

The theoretical lineage of this model draws from D’orner’s PSI theory (hydraulic-like needs such as competence, certainty, affiliation) and Scherer’s Component Process Model of emotion (affect as a continuously updating synchronization of subsystems).

The core thesis is that intent is cheap, but expression is expensive. Transforming a high-level goal (e.g., “write a report”) into a specific sequence of tokens requires the expenditure of cognitive resources. When these resources are depleted—manifesting as low Energy (E) or Patience (P)—the agent does not necessarily stop working; rather, the quality and nature of the work shifts.

3 The State Vector $S(t)$: Dimensional Analysis and Cognitive Substrates

The cognitive state is defined as

$$S(t) = [E(t), N(t), P(t), A(t)]^T \in [0, 1]^4. \quad (1)$$

These four dimensions map to fundamental neurobiological systems and well-established constructs in cognitive psychology.

3.1 Energy (E): The Metabolic Governor

Energy (E) represents the agent’s capacity for sustained cognitive effort.

- Cognitive function. High E supports deep, recursive reasoning and extensive forward planning; low E forces heuristic, “lossy” processing.
- Biological analog. Correlates with glucose availability, glycogen stores in astrocytes, and accumulation of adenosine.
- Conservation behavior. As $E \rightarrow 0$, the agent conserves resources by increasing decision thresholds and reducing verbosity.

3.2 Novelty-Seeking (N): The Dopaminergic Drive

Novelty-seeking (N) quantifies openness to new stimuli and propensity for distraction.

- Cognitive function. High N lowers barriers to switching contexts, making the agent more likely to abandon a current task for a novel stimulus.
- Biological analog. Models tonic/phasic dopaminergic activity; correlates with exploratory behavior and creativity but reduces sustained focus on repetitive tasks.

3.3 Patience (P): Executive Control and Inhibition

Patience (P) measures inhibitory control and tolerance for delay or complexity.

- DDM interpretation. In a Drift Diffusion Model (DDM), P maps to boundary separation a (caution): higher P reduces errors but increases latency.
- Biological analog. Reflects prefrontal cortex integrity; fatigue and stress degrade PFC function and increase impulsivity.
- Caffeine interaction. Caffeine may increase arousal yet reduce patience, producing “fast errors”.

3.4 Arousal (A): The Activation System

Arousal (A) represents emotional reactivity and physiological activation.

- Cognitive function. Moderate A can improve performance (Yerkes–Dodson), while extreme A causes attentional tunneling and reduced working memory.
- Biological analog. Maps to LC–NE (locus coeruleus–norepinephrine) and HPA-axis dynamics.

4 Temporal Dynamics and Circadian Regulation

Between events, the state follows mean-reverting stochastic dynamics toward a time-varying circadian attractor $S^*(t)$:

$$\frac{dS}{dt} = -\Lambda (S(t) - S^*(t)) + \eta(t), \quad \eta(t) \sim \mathcal{N}(0, \Sigma), \quad (2)$$

with

$$\Lambda = \text{diag}(\lambda_E, \lambda_N, \lambda_P, \lambda_A). \quad (3)$$

Table 1: Example relaxation rates and qualitative physiological basis.

State	Rate (min^{-1})	Half-life (min)	Basis (qualitative)
Energy λ_E	0.02	34.6	Recovery from acute cognitive fatigue is relatively slow; partial re
Novelty λ_N	0.05	13.9	Habituation to novelty tends to be faster with
Patience λ_P	0.03	23.1	Executive control replenishment is moderate and
Arousal λ_A	0.10	6.9	Fast catecholamine clearance plus a slower tail f

4.1 Calibration of relaxation rates (Λ)

The relatively rapid decay of arousal compared to energy yields a common hysteresis: an agent can calm down (low A) long before it recovers its capacity for sustained work (high E).

4.2 The circadian attractor $S^*(t)$

The baseline attractor is modeled as an empirically tuned schedule (interpolate with a smooth spline between points):

Table 2: Circadian attractor schedule $S^*(t)$ (example).

Time window	E^*	N^*	P^*	A^*	Phenotype
08:00–10:00	0.85	0.30	0.70	0.40	Morning peak: precise, formal, deep work.
12:00–14:00	0.25	0.20	0.30	0.25	The dip: low engagement, slow latency, “silence”.
14:00–17:00	0.75	0.45	0.65	0.50	Second wind: productive, moderate distractions.
18:00–22:00	0.25	0.75	0.30	0.55	Evening mode: casual, novelty-seeking, reactive.
22:00–08:00	0.05	0.10	0.10	0.10	Sleep: high thresholds, minimal activity.

5 Illustrative Figures

6 Event-Driven Discontinuities: The Impulse Model

The system is a hybrid dynamical system: continuous flow interrupted by discrete jumps.

6.1 Stimulus impulse $\Delta S_{\text{stimulus}}$

When a stimulus arrives,

$$S(t^+) = S(t^-) + \Delta S_{\text{stimulus}}, \quad \Delta S_{\text{stimulus}} = \begin{bmatrix} 0 \\ \sigma_N \nu \\ -\sigma_P c \\ \sigma_A s \end{bmatrix}, \quad (4)$$

where s is salience, ν is novelty, and c is complexity.

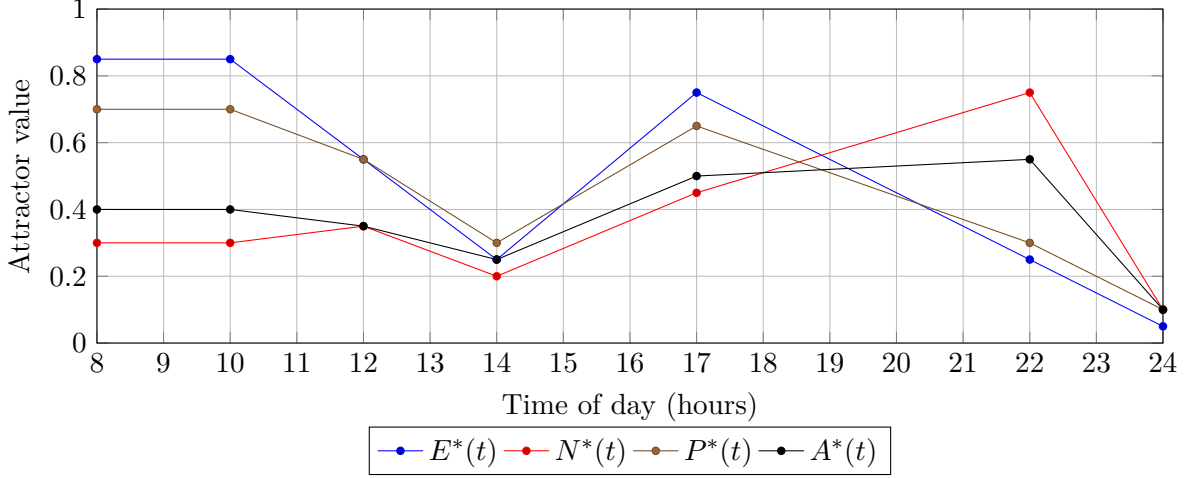


Figure 1: Example circadian attractors (anchor points; connect with a smooth spline in an implementation).

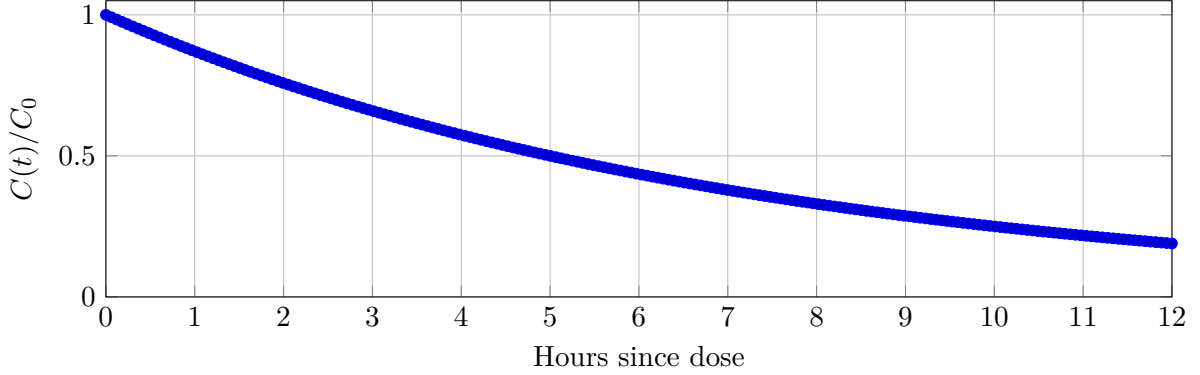


Figure 2: Simplified caffeine elimination curve with half-life $\tau_{\text{half}} = 5$ h (normalised).

6.2 Action impulse ΔS_{action}

When the agent acts,

$$S \rightarrow S + \Delta S_{\text{action}}, \quad \Delta S_{\text{action}} = \begin{bmatrix} -\delta_E \\ \delta_N \nu \\ -\delta_P \\ \delta_A \epsilon \end{bmatrix}, \quad (5)$$

where $\epsilon \in [0, 1]$ is emotional valence/intensity.

7 Exogenous Modulation: The Pharmacokinetics of Caffeine

7.1 Pharmacokinetic (PK) model

A two-rate absorption/elimination form is:

$$C(t) = \text{Dose} \frac{k_a}{k_a - k_e} \left(e^{-k_e t} - e^{-k_a t} \right). \quad (6)$$

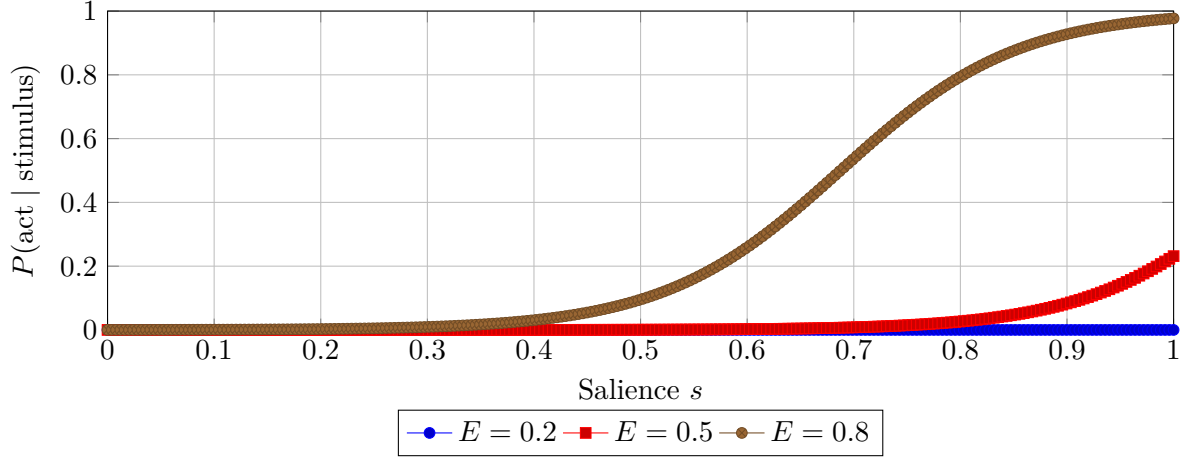


Figure 3: Example adaptive action probability curves under $\theta(t) = \theta_0/E(t)$ (illustrative only).

For a simplified elimination-only approximation after absorption:

$$C(t) = C_0 \exp\left(-\frac{t}{\tau_{\text{half}}}\right), \quad \tau_{\text{half}} \approx 5 \text{ hours.} \quad (7)$$

7.2 Pharmacodynamic (PD) modulation of attractors

Caffeine modulates the circadian attractors as

$$E^*(t) \leftarrow E^*(t) + k_E C(t), \quad (8)$$

$$A^*(t) \leftarrow A^*(t) + k_A C(t), \quad (9)$$

$$P^*(t) \leftarrow P^*(t) - k_P C(t). \quad (10)$$

The characteristic “jitter” signature corresponds to high E , high A , and low P .

8 Decision Architecture: Thresholds and Latency

8.1 Action probability and adaptive thresholding

Define a dynamic response threshold

$$\theta(t) = \frac{\theta_0}{E(t) + \alpha N(t) \nu}. \quad (11)$$

Action probability is then

$$P(\text{act} \mid \text{stimulus}) = \sigma(s - \theta(t)) = \frac{1}{1 + \exp(-k(s - \theta(t)))}. \quad (12)$$

8.2 Drift diffusion model (DDM) mapping for latency

Rather than sampling latency directly from a fixed distribution, map cognitive state into DDM parameters:

- Drift rate $v \propto E(t)$ (faster evidence accumulation with higher energy).

- Boundary separation $a \propto P(t)$ (more caution with higher patience).
- Non-decision time $T_{er} \propto 1/A(t)$ (faster encoding/motor prep under higher arousal).

Table 3: DDM parameter mapping (core relationships).

DDM parameter	State component	Relationship	Effect of high state
Drift v	Energy E	$v \propto E$	Lower latency via faster accumulation.
Boundary a	Patience P	$a \propto P$	Higher accuracy but slower responses.
Non-decision T_{er}	Arousal A	$T_{er} \propto 1/A$	Faster encoding/motor response.
Starting point z	Bias (optional)	—	Captures expectation/priors as an extension.

9 Expression Modulation: Psycholinguistics of State

Define generation modifiers:

9.1 Verbosity

$$V(t) = V_{\text{base}} E(t)^{\gamma_V} P(t)^{\delta_V}. \quad (13)$$

9.2 Formality / tone

$$F(t) = F_{\text{base}} (1 - N(t)) (1 - A(t)). \quad (14)$$

9.3 Depth

$$D(t) = D_{\text{base}} E(t) P(t). \quad (15)$$

9.4 Error rate

$$\epsilon_{\text{error}}(t) = \frac{\epsilon_{\text{base}}}{E(t) P(t)}. \quad (16)$$

10 Observable Signatures and Forensic Validation

11 Implementation Architecture: The “Sidecar” Loop

11.1 Data structures

```
interface CognitiveState {
  E: number; // Energy [0,1]
  N: number; // Novelty [0,1]
  P: number; // Patience [0,1]
  A: number; // Arousal [0,1]
```

Table 4: Qualitative signatures produced by the model.

Behavioral pattern	State mechanism	Human correlate (qualitative)
Morning bursts	High E , high P , low N	Peak typing speed/accuracy late morning.
Lunch lull	Trough in E^* and P^*	Post-lunch dip in vigilance tasks.
Evening “hot takes”	Low E , high N , high A	Reduced inhibition and increased novelty-seeking.
Burst-then-quiet	Action impulse depletes E	Refractory periods after intense effort.
Caffeine jitters	High E , high A , low P	Impulsivity and higher error rate at high dose.

```
t: number; // Last update timestamp
caffeine_mg: number; // Active caffeine load
}
```

11.2 Update loop (schematic)

```
EVERY Dt (or on stimulus arrival):

1. Evolve toward  $S^*(t)$ :  $dS/dt = -\text{Lambda}*(S - S^*(t)) + \text{noise}$ 
2. Update caffeine load:  $\text{caffeine\_mg} \leftarrow \text{caffeine\_mg} * \exp(-Dt/\text{tau\_half})$ 
3. Modulate attractors:  $E^*, A^*, P^* \leftarrow E + kE*C, A + kA*C, P - kP*C$ 
4. If stimulus pending:
  a. Compute salience  $s$ , novelty  $nu$ , complexity  $c$ 
  b. Apply stimulus impulse
  c. Compute threshold  $\text{theta}(t)$ 
  d. Sample  $\text{act} \sim \text{Bernoulli}(\text{sigmoid}(s - \text{theta}))$ 
5. If act:
  a. Derive DDM parameters  $(v, a, \text{Ter})$  from  $(E, P, A)$ 
  b. Sample latency  $\text{tau}$  and wait  $\text{tau}$ 
  c. Compute modifiers  $(V, F, D, \text{eps\_error})$ 
  d. Generate response under modifiers
  e. Apply action impulse
6. Persist state
```

12 Conclusion

The Caffeine Algorithm offers a biologically grounded framework for agent behavior by formalizing circadian baselines, homeostatic depletion, and exogenous modulation. Realism emerges not by adding capabilities, but by simulating constraints.

13 References (informal)

- Yerkes–Dodson (1908) — Arousal-performance curve.
- Borbély (1982) — Two-process model of sleep regulation.
- Kahneman (1973) — Attention and Effort.
- Hockey (1997) — Compensatory control in performance regulation.
- Mark et al. (2008) — The cost of interrupted work.
- Csikszentmihalyi (1990) — Flow states.
- Dörner — PSI theory (need-driven regulation).
- Scherer — Component Process Model of emotion.